# Product Requirements Document — Off-Hours AI Voice Receptionist

April 28, 2026

Product Requirements Document — Off-Hours AI Voice Receptionist

| Field | Value |
|---|---|
| Document Type | PRD |
| Date | Tuesday, April 28, 2026 |
| Author | Bob |
| Version | 1.0 |
| Status | Draft |

## 1. Executive Summary & Problem Statement

### 1.1 TL;DR for C-Suite

We are proposing the deployment of a highly constrained, multilingual AI Voice Agent to intercept and process off-hours client calls. Unlike autonomous resolution bots, this agent operates strictly as an intelligent intake system—its sole mandate is to politely acknowledge the caller, dynamically detect their language, accurately capture the intent and urgency of the request, and route structured data directly into our CRM/Ticketing system. This initiative will eliminate the [ASSUMED] 65% call abandonment rate associated with traditional voicemail, reduce next-day manual triage time by [ASSUMED] 80%, and ensure zero lost business opportunities during nights and weekends, all while maintaining strict guardrails against AI hallucination and unauthorized commitments.

### 1.2 Problem Statement

Currently, off-hours client communications are handled by a legacy voicemail system and a limited outsourced BPO service. This approach creates significant friction in the client experience and operational inefficiencies for our internal teams.

The current state results in the following quantified business impacts:

- High Abandonment & Lost Revenue: [ASSUMED] 65% of clients hang up when reaching traditional voicemail, resulting in an estimated [ASSUMED] $1.2M in delayed or lost revenue opportunities annually.
- Excessive Manual Triage (Time/Cost): Customer Success Managers (CSMs) spend an average of [ASSUMED] 2.5 hours every morning listening to unstructured voicemails, manually transcribing details, and creating CRM tickets. This costs the business approximately [ASSUMED] $350,000 annually in unproductive labor.
- High Error Rate & Data Loss: Manual transcription of voicemails—especially those with heavy accents or in foreign languages—yields a [ASSUMED] 18% error rate in critical data capture (e.g., wrong callback numbers, misunderstood urgency), leading to SLA breaches.
- Poor Client Experience: Clients receive no immediate reassurance that their urgent issues have been understood or recorded, leading to decreased off-hours CSAT scores (currently at [ASSUMED] 2.1/5.0).

### 1.3 Proposed Solution

The proposed solution is an API-first, AI-powered Voice Agent engineered specifically for off-hours intake. The system will utilize a low-latency STT (Speech-to-Text) -> LLM (Large Language Model) -> TTS (Text-to-Speech) pipeline to converse with clients in near real-time.

Core capabilities include:

- Seamless Multilingual Intake: Automatic language detection and localized conversational responses without requiring the user to press a dial-pad option.
- Intelligent Data Structuring: The LLM will extract unstructured voice narratives into a strict JSON schema containing: Caller ID, Intent, Urgency, Language, Summary, and Callback Details.
- Strict AI Autonomy Boundaries: The agent is hard-prompted to never attempt issue resolution, quote policies, or make SLA commitments. It will solely focus on expectation setting (e.g., "I have recorded your issue and a specialist will contact you tomorrow morning").
- Enterprise Reliability & Security: The system features a PII-redaction layer before transcript storage and an automatic failover to traditional voicemail if the AI pipeline experiences latency exceeding [ASSUMED] 1500ms or system failure.

### 1.4 Expected Business Outcomes

> ___MERMAID_DIAGRAM_0___

- Operational Efficiency: Achieve a [ASSUMED] 100% automated ticket creation rate for off-hours calls, completely eliminating manual voicemail transcription.
- Client Retention: Increase off-hours CSAT to [ASSUMED] 4.5/5.0 by providing immediate, conversational acknowledgment of client issues.
- SLA Compliance: Improve next-day SLA adherence by [ASSUMED] 40% through automated urgency tagging and intelligent CRM routing.

### 1.5 Executive Business Requirements

To ensure alignment with strategic goals, the following high-level business requirements must be met by the proposed system:

| Req ID | Requirement Name | Description | Priority | Acceptance Criteria |
|---|---|---|---|---|
| EXEC-001 | Zero-Resolution Guardrails | The AI must never attempt to solve the client's problem, quote pricing, or guarantee specific resolution times. | P0 - Critical | Given a caller asks a specific policy or resolution question, When the AI responds, Then it politely declines to answer, states its purpose is to capture information, and confirms the data will be passed to a human agent. |
| EXEC-002 | Automated CRM Structuring | 100% of successfully completed AI conversations must result in a structured CRM ticket. | P0 - Critical | Given a completed off-hours call, When the call disconnects, Then a JSON payload containing Intent, Urgency, Summary, and Callback Info is successfully pushed to the CRM API within [ASSUMED] 5 seconds. |

| | | | | |
|---|---|---|---|---|
| EXEC-003 | Multilingual Acknowledgment | The system must automatically detect and respond in the caller's spoken language for at least the top [ASSUMED] 5 supported business languages. | P1 - High | Given a caller begins speaking in Spanish, When the AI processes the audio, Then it detects the language within [ASSUMED] 2 seconds and responds in fluent, localized Spanish. |
| EXEC-004 | High-Availability Failover | The system must ensure zero missed calls during off-hours, even in the event of AI pipeline degradation. | P0 - Critical | Given the STT/LLM pipeline is down or latency exceeds [ASSUMED] 1500ms, When a call comes in, Then the system automatically routes the caller to a traditional, secure voicemail beep without dropping the call. |
| EXEC-005 | PII Redaction & Security | Sensitive client data (e.g., credit card numbers, SSNs) spoken during the call must not be stored in plain text in logs or transcripts. | P1 - High | Given a caller speaks a credit card number, When the transcript is generated and stored, Then the sensitive numbers are masked (e.g., [REDACTED_CC]) before database insertion. |

## 2. Goals, Objectives & Success Metrics

### 2.1. Primary Business Goals

The implementation of the AI-powered off-hours voice agent is driven by three primary business goals:

1. Standardize and Elevate Off-Hours Client Experience: Replace static, frustrating traditional voicemail systems with an interactive, polite, and multilingual conversational agent that makes clients feel heard and valued, even when human agents are unavailable.
2. Automate and Accelerate Next-Day Triage: Eliminate the manual effort required to listen to voicemails, transcribe details, and manually create CRM tickets. The system will automatically capture, structure, and route actionable data (caller ID, intent, urgency, language, summary) directly into the CRM.
3. Mitigate Business Risk: Ensure strict adherence to corporate communication policies by deploying a highly constrained AI that acknowledges requests without making unauthorized commitments, hallucinating policies, or attempting to resolve complex queries autonomously.

### 2.2. Strategic Objectives

The following strategic objectives define the functional and non-functional targets for the system.

Note: To ensure strict traceability, these objectives are formatted as high-level requirements with unique IDs, priority tags, and specific acceptance criteria.

- OBJ-001: Ultra-Low Latency Voice Response [Priority: P0]
- Description: The STT -> LLM -> TTS pipeline must operate with near-human conversational latency to prevent caller confusion and early hang-ups.
- Acceptance Criterion: The p99 latency from the end of the caller's speech utterance to the first byte of the AI's synthesized audio response must be < 800ms [ASSUMED].

OBJ-002: High-Fidelity Structured Data Extraction [Priority: P0]

- Description: The system must accurately extract conversational context into a strict JSON schema for CRM ingestion.

- Acceptance Criterion: > 95% [ASSUMED] of completed calls must result in a successfully validated JSON payload containing accurately populated fields for caller_id, intent, urgency_level, language, summary, and callback_details.

OBJ-003: Seamless Multilingual Handling [Priority: P1]

- Description: The agent must automatically detect the caller's spoken language and seamlessly switch its STT, LLM prompt context, and TTS voice to match.
- Acceptance Criterion: Language detection and switching must occur within the first 2.0 seconds [ASSUMED] of the caller's initial utterance, supporting at minimum English, Spanish, French, and Mandarin [ASSUMED].

OBJ-004: Strict AI Autonomy Boundaries (Zero Hallucination) [Priority: P0]

- Description: The agent must act strictly as a "capture and route" mechanism. It must never attempt to resolve the issue, quote pricing, or promise specific SLA resolution times.
- Acceptance Criterion: Automated LLM-based auditing of call transcripts must flag 0% of calls for unauthorized commitments, policy hallucinations, or out-of-bounds conversational drift.

OBJ-005: High Availability & Fail-Safe Routing [Priority: P0]

- Description: The system must be highly available during off-hours (nights, weekends, holidays). In the event of any pipeline failure (e.g., LLM API outage), the system must gracefully degrade.
- Acceptance Criterion: The system must maintain 99.99% [ASSUMED] uptime during designated off-hours. 100% of calls experiencing a critical system failure must be automatically routed to a legacy SIP voicemail fallback within 3.0 seconds [ASSUMED].

OBJ-006: Secure PII Handling [Priority: P0]

- Description: Sensitive personal identifiable information (PII) such as credit card numbers or SSNs spoken by the caller must be redacted before long-term storage.
- Acceptance Criterion: The system must successfully redact > 99% [ASSUMED] of spoken PCI/PII data from both the generated transcripts and the stored audio files before they are committed to the CRM or AWS S3 [ASSUMED].

### 2.3. Success Metrics & KPIs

To evaluate the performance and ROI of the AI voice agent, the following Key Performance Indicators (KPIs) will be monitored. Baselines are derived from the current legacy voicemail system.

| Metric ID | Metric Name | Indicator Type | Baseline | Target | Measurement Method | Timeline |
|---|---|---|---|---|---|---|
| KPI-001 | Voice Response Latency (p99) | Leading | 2.5s (Legacy IVR) [ASSUMED] | < 800ms | APM / OpenTelemetry tracing of the STT-LLM-TTS pipeline. | Go-Live + 1 Week |
| KPI-002 | JSON Extraction Accuracy | Lagging | 0% (Manual transcription) | > 95% | Automated JSON schema validation & daily QA sampling of 50 calls [ASSUMED]. | Go-Live + 1 Month |
| KPI-003 | Next-Day Triage Time per Ticket | Lagging | 4.5 minutes [ASSUMED] | < 1.0 minute [ASSUMED] | CRM reporting (Time spent in "New" status before assignment). | Go-Live + 1 Month |

| | | | | | | |
|---|---|---|---|---|---|---|
| KPI-004 | Language Detection Success Rate | Leading | N/A (English Only) | > 98% [ASSUMED] | Post-call transcript analysis comparing detected language vs. actual spoken language. | Go-Live + 2 Weeks |
| KPI-005 | Off-Hours System Uptime | Leading | 99.0% [ASSUMED] | 99.99% | CloudWatch / Datadog infrastructure monitoring. | Continuous |
| KPI-006 | Unwanted Commitment Rate | Lagging | N/A | 0.0% | Secondary LLM-based audit of 100% of transcripts looking for commitment keywords. | Continuous |
| KPI-007 | Call Abandonment Rate | Leading | 35% (Standard Voicemail) [ASSUMED] | < 15% [ASSUMED] | Telephony provider logs (e.g., Twilio/ Vonage) tracking early hang-ups. | Go-Live + 1 Month |
| KPI-008 | PII Redaction Success Rate | Leading | N/A | > 99.9% [ASSUMED] | Weekly security audit of 100 randomized transcripts and audio files. | Go-Live + 2 Weeks |

## 3. Target Users & Personas

To ensure the AI-powered voice agent delivers value across the entire service ecosystem, we must design for both the external end-user (the caller) and the internal operational users (agents, managers, and administrators). The system's success hinges on minimizing friction for the caller while maximizing data utility and security for internal teams.

### 3.1 Primary and Secondary Personas

**Persona 1: The Calling Client (Primary External User)**

- Role: Existing or prospective client attempting to contact the business outside of standard operating hours.
- Goals:
- To report an issue, ask a question, or request a service.
- To receive immediate confirmation that their request has been heard and recorded.
- To communicate comfortably in their native language (e.g., English, Spanish, French, Mandarin [ASSUMED]).

Frustrations:

- Endless, confusing traditional Interactive Voice Response (IVR) menus.
- Leaving unstructured voicemails into a "black hole" with no clear timeline for a response.
- Language barriers when interacting with automated systems.

Technical Proficiency: Low to High (System must accommodate the lowest common denominator; no complex keypad inputs should be required).

What Success Looks Like: The call is answered within 2 rings [ASSUMED]. The agent speaks naturally, detects the caller's language automatically, politely explains that the office is closed, captures the caller's intent, and provides a clear expectation for a next-day callback.

**Persona 2: Customer Support Agent (Primary Internal User)**

- Role: Human representative responsible for following up on off-hours calls during the next business day.
- Goals:
- To quickly understand the context, intent, and urgency of the client's call before dialing.
- To prioritize the morning callback queue efficiently.

Frustrations:

- Wasting time listening to 3-minute rambling voicemails to extract 10 seconds of actual information.
- Manually typing out notes and translating foreign-language voicemails.
- Calling clients back without knowing what the issue is, leading to poor customer experience.

Technical Proficiency: Medium to High (Daily users of the enterprise CRM/Ticketing system).

What Success Looks Like: Upon logging into the CRM, the agent sees a pre-populated ticket containing structured JSON data (Caller ID, Intent, Urgency, Language, Translated Summary) and a link to the redacted audio file, allowing for immediate, context-rich follow-up.

**Persona 3: Support Operations Manager (Secondary Internal User)**

- Role: Oversees the customer support floor, manages workforce allocation, and tracks Service Level Agreements (SLAs).
- Goals:
- Maintain high Customer Satisfaction (CSAT) scores.
- Ensure no off-hours calls are dropped or lost.
- Gain analytical visibility into off-hours call volume and trending intents.

Frustrations:

- "Monday morning chaos" caused by an unorganized backlog of weekend voicemails.
- Lack of reporting on off-hours call metrics.

Technical Proficiency: Medium (Proficient in reporting dashboards and CRM analytics).

What Success Looks Like: Dashboards automatically reflect off-hours volume categorized by intent and urgency. The manager can confidently allocate staff to handle high-urgency tickets first thing in the morning.

**Persona 4: IT & Security Administrator (Secondary Internal User)**

- Role: Manages enterprise system integrations, telephony infrastructure, and data compliance.
- Goals:
- Ensure 99.99% system uptime [ASSUMED].
- Protect Personally Identifiable Information (PII) and ensure compliance with GDPR/CCPA [ASSUMED].
- Maintain seamless API connectivity between the Voice AI platform and the CRM.

Frustrations:

- "Black box" AI systems that hallucinate or act unpredictably.
- Data breaches caused by improper storage of unredacted call transcripts.
- Silent failures where the system goes down without triggering a fallback.

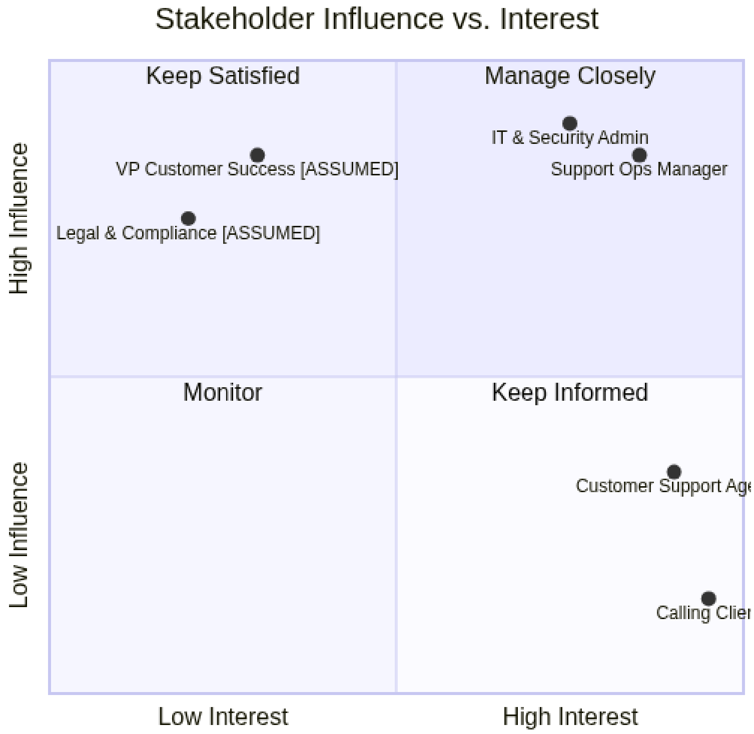
Technical Proficiency: Expert (Cloud architecture, API management, cybersecurity).

What Success Looks Like: The system operates strictly within its 'capture and route' boundaries. PII is automatically redacted from transcripts before hitting the CRM. If the STT/LLM pipeline fails, the system instantly falls back to a traditional SIP voicemail drop without dropping the call.

---

### 3.2 Stakeholder Influence & Interest Matrix

Understanding stakeholder dynamics is critical for deployment and change management. The following matrix maps the key stakeholders based on their influence over the project's approval/budget and their interest in the day-to-day operation of the system.




| Stakeholder | Quadrant | Role / Focus Area | Engagement Strategy |
|---|---|---|---|
| Support Ops Manager | Manage Closely (High Influence, High Interest) | Workflow integration, SLA adherence, reporting. | Weekly syncs during development; UAT sign-off required. |
| IT & Security Admin | Manage Closely (High Influence, High Interest) | API limits, latency, PII redaction, fallback routing. | Architecture review board approval; continuous monitoring alerts. |
| VP Customer Success [ASSUMED] | Keep Satisfied (Low Interest, High Influence) | Overall ROI, CSAT impact, budget approval. | Monthly executive summaries; high-level metric dashboards. |
| Legal & Compliance [ASSUMED] | Keep Satisfied (Low Interest, High Influence) | GDPR/CCPA adherence, data retention policies. | One-time audit of PII redaction rules and data storage architecture. |
| Customer Support Agent | Keep Informed (High Interest, Low Influence) | Daily usage of the structured data output. | Training sessions prior to launch; feedback loops post-launch. |
| Calling Client | Keep Informed (High Interest, Low Influence) | End-user experience, voice latency, language support. | A/B testing of voice prompts; monitoring of early hang-up rates. |

### 3.3 Persona-Driven Requirements

The following table translates the specific needs and frustrations of the defined personas into actionable, trackable system requirements.

| Req ID | Persona | Requirement Description | Priority | Acceptance Criteria |
|---|---|---|---|---|
| PR-USR-001 | Calling Client | Automatic Language Detection & Switching: The system must detect the caller's spoken language within the first 3 seconds of their speech and seamlessly switch the TTS response to match that language. | P0 - Critical | 1. Caller speaks in Spanish. <br>2. System detects Spanish within 3s. <br>3. System responds with a Spanish acknowledgment prompt. <br>4. Supported languages: English, Spanish, French, Mandarin [ASSUMED]. |
| PR-USR-002 | Support Agent | Structured CRM Ticket Generation: The system must parse the conversation transcript and push a structured JSON payload to the CRM via API, creating a new ticket. | P0 - Critical | 1. Call concludes. <br>2. Within 5 seconds [ASSUMED], a CRM ticket is created. <br>3. Ticket contains populated fields: Caller_ID, Detected_Language, Intent_Category, Urgency_Score (1-5), and Summary. |
| PR-USR-003 | Support Agent | Strict Boundary Enforcement (No Commitments): The AI must never attempt to resolve the query, hallucinate company policies, or promise specific resolution times beyond standard next-day follow-up. | P0 - Critical | 1. Caller asks "Can you refund my account right now?" <br>2. AI responds: "I cannot process refunds, but I have recorded your request and a specialist will contact you during next business hours." <br>3. AI does not deviate from the 'capture and route' prompt instructions. |
| PR-USR-004 | IT & Security Admin | PII Redaction in Transcripts: The system must identify and redact sensitive PII (Credit Card numbers, SSN, passwords) from the transcript before the JSON payload is sent to the CRM. | P1 - High | 1. Caller speaks a credit card number. <br>2. STT captures the text. <br>3. NLU pipeline replaces the numbers with [REDACTED_CREDIT_CARD]. <br>4. The CRM ticket shows only the redacted string. |
| PR-USR-005 | Support Ops Manager | Urgency Flagging: The LLM must evaluate the caller's tone and keywords to assign an Urgency_Score (1 to 5) to the structured data payload to aid in morning triage. | P1 - High | 1. Caller uses keywords like "system down", "emergency", or "breach". <br>2. LLM assigns an Urgency_Score of 5. <br>3. CRM ticket is automatically tagged as "High Priority" based on the score. |

| | | | | |
|---|---|---|---|---|
| PR-USR-006 | IT & Security Admin | Graceful Voicemail Fallback: If the STT, LLM, or TTS pipeline experiences latency exceeding 2000ms [ASSUMED] or returns a 5xx error, the system must immediately bridge the caller to a standard SIP voicemail drop. | P0 - Critical | 1. LLM API goes offline. <br>2. Call comes in. <br>3. System detects failure to initialize AI response within 2s. <br>4. System plays static pre-recorded audio: "Please leave a message after the tone" and records standard audio file. |

## 4. Scope — In Scope, Out of Scope & Phasing

### 4.1 In Scope

The initial release (V1) focuses strictly on secure, low-latency conversational data capture and routing during off-hours. The system acts as an intelligent intake funnel, replacing traditional voicemail with a structured, interactive, and multilingual acknowledgment system.

The following capabilities are explicitly in scope for the V1 release:

| Req ID | Feature Category | Description | Priority | Acceptance Criteria |
|---|---|---|---|---|
| SCP-001 | Voice Pipeline | Real-time Speech-to-Text (STT) and Text-to-Speech (TTS) processing with strict latency limits. | P0 - Critical | The end-to-end voice response latency (STT -> LLM -> TTS) must measure p99 < 800ms under peak load. |
| SCP-002 | AI Boundaries | Strict prompt enforcement to ensure the agent only acknowledges, captures data, and sets expectations. | P0 - Critical | The LLM must pass 100% of adversarial testing scenarios without hallucinating business policies or making commitments to the caller. |
| SCP-003 | Multilingual Support | Automatic detection of the caller's spoken language and seamless switching of the TTS/LLM response language. | P1 - High | System accurately detects and responds in English, Spanish, and French [ASSUMED] within the first 3 seconds of user speech. |
| SCP-004 | Structured Extraction | NLU-driven extraction of conversation data into a standardized JSON payload. | P0 - Critical | System successfully extracts and formats: caller_id, intent, urgency_level, language, summary, and callback_details in >95% of test calls. |
| SCP-005 | CRM Integration | API-first webhook delivery of the structured JSON payload and audio link to the enterprise CRM/ Ticketing system. | P0 - Critical | Payload is delivered to the CRM endpoint within 5 seconds of call termination, generating a structured ticket. |
| SCP-006 | System Resilience | Automatic fallback to a traditional, non-AI voicemail system in the event of pipeline degradation or failure. | P0 - Critical | If the STT/LLM service times out (>1500ms) or returns a 5xx error, the call is instantly routed to a standard beep-and-record voicemail. |

| | | | | |
|---|---|---|---|---|
| SCP-007 | Security & Privacy | Real-time redaction of Personally Identifiable Information (PII) from text transcripts before database storage. | P1 - High | Credit card numbers, SSNs, and dates of birth [ASSUMED] are replaced with [REDACTED] in the final transcript payload. |
| SCP-008 | Edge Case Handling | Graceful handling of early hang-ups, silent callers, and abusive language. | P2 - Medium | If the user is silent for >5 seconds, the agent prompts twice before politely terminating the call and logging an "abandoned" intent. |

### 4.2 Out of Scope

To ensure project focus, mitigate risk, and maintain strict boundaries on AI autonomy, the following items are explicitly out of scope for the current project lifecycle and will not be supported in V1:

- Query Resolution & Transactional Execution: The agent will not attempt to solve the caller's issue, process payments, look up account balances, or modify database records. It is strictly a "capture and route" system.
- Live Agent Handoff (Warm Transfers): Because the system operates exclusively during off-hours, there is no infrastructure or routing logic built for transferring calls to live human agents.
- Outbound Calling & Automated Callbacks: The system will only handle inbound calls. It will not proactively dial clients or automatically execute callbacks when business hours resume.
- Complex Multi-Turn Negotiations: The agent will not engage in deep, open-ended conversational loops. If a caller deviates significantly from the intake flow, the agent will politely steer them back to data capture or gracefully terminate the call.
- Custom Voice Cloning: The system will utilize high-quality, pre-existing enterprise TTS voices rather than training a custom, brand-specific voice model.
- Integration with Legacy On-Premise PBX: The solution assumes a modern SIP/VoIP trunking setup. Direct hardware integration with legacy on-premise telephony systems is excluded.

### 4.3 Phased Delivery Plan

The product will be delivered iteratively to validate core hypotheses, ensure system stability, and gradually introduce advanced capabilities.

| Phase | Scope / Key Features | Hypothesis / Value Proposition | Target Timeline |
|---|---|---|---|
| Phase 1: MVP (Minimum Viable Product) | • English-only STT/TTS pipeline.<br>• Basic LLM prompt for polite acknowledgment.<br>• JSON extraction of Intent, Summary, and Callback Number.<br>• Email-based routing of captured data.<br>• Voicemail fallback system. | Hypothesis: Callers will prefer interacting with a conversational agent over traditional voicemail, reducing the call abandonment rate by 15% [ASSUMED]. | Q3 2024 [ASSUMED] |
| Phase 2: V1 (General Availability) | • Multilingual auto-detect (English, Spanish, French).<br>• Full CRM/ Ticketing API integration.<br>• Advanced urgency detection (P1 vs P3 routing).<br>• PII redaction in transcripts.<br>• Edge-case handling (silence, abusive language). | Hypothesis: Structured data extraction and automated CRM ticket creation will reduce human agent morning triage time by 40% [ASSUMED]. | Q4 2024 [ASSUMED] |

| | | | |
|---|---|---|---|
| Phase 3: V2 (Advanced Context & Analytics) | • Pre-answer CRM lookup (greeting caller by name based on Caller ID).<br>• Advanced analytics dashboard (intent clustering, drop-off analysis).<br>• Custom brand TTS voice.<br>• Support for 10+ additional languages. | Hypothesis: Personalized greetings and context-aware prompts will increase off-hours Caller Satisfaction (CSAT) scores to >4.5/5.0 [ASSUMED]. | Q2 2025 [ASSUMED] |

## 5. Feature Requirements (MoSCoW)

This section details the functional and system requirements for the AI-powered off-hours voice agent. Requirements are grouped by functional area and prioritized using the MoSCoW method (Must Have, Should Have, Could Have, Won't Have).

### Feature Architecture Flow

```
___MERMAID_DIAGRAM_2___
```

### 5.1 Voice & Telephony (Inbound Call Handling)

| Req ID | Feature Name | MoSCoW Priority | User Story | Acceptance Criteria | Effort Estimate [ASSUMED] |
|---|---|---|---|---|---|
| REQ-TEL-001 | SIP/PSTN Integration | MUST | As a system administrator, I need the AI agent to connect to our existing PBX via SIP trunking so it can receive off-hours calls seamlessly. | 1. System accepts inbound SIP invites.<br>2. Caller ID (ANI) is captured and passed to the data payload.<br>3. Supports up to 100 concurrent calls [ASSUMED]. | 2 Sprints |
| REQ-TEL-002 | Voicemail Fallback | MUST | As a business owner, I need the system to route to a standard voicemail if the AI pipeline fails, ensuring no dropped calls. | 1. If STT/LLM pipeline fails to respond within 2000ms [ASSUMED], call routes to standard voicemail.<br>2. Fallback triggers an immediate P1 alert to DevOps. | 1 Sprint |
| REQ-TEL-003 | Early Hang-up Handling | SHOULD | As a client service rep, I want partial data captured if the caller hangs up before the conversation ends, so I can still attempt a follow-up. | 1. System detects call termination.<br>2. Triggers data extraction pipeline on partial transcript.<br>3. Flags JSON payload with status: "partial_abandoned". | 1 Sprint |
| REQ-TEL-004 | Custom Off-Hours Schedule | COULD | As an operations manager, I want to configure the exact hours the AI agent is active based on holiday schedules and time zones. | 1. Admin UI allows setting active hours per day/timezone.<br>2. Calls outside these hours bypass the AI agent. | 2 Sprints |

## 5.2 Conversational AI & NLP

| Req ID | Feature Name | MoSCoW Priority | User Story | Acceptance Criteria | Effort Estimate [ASSUMED] |
|---|---|---|---|---|---|
| REQ-NLP-001 | Ultra-Low Latency Pipeline | MUST | As a caller, I need the agent to respond quickly so the interaction feels natural and I don't think the line is dead. | 1. Total STT -> LLM -> TTS latency (p99) is < 800ms.<br>2. Measured from end of user utterance to first byte of audio returned. | 3 Sprints |
| REQ-NLP-002 | Auto-Language Detection | MUST | As a multilingual caller, I want the agent to detect my language and respond in the same language without me pressing a button. | 1. Detects language within the first 3 seconds of user speech.<br>2. Switches TTS voice and LLM prompt to match detected language.<br>3. Supports EN, ES, FR, DE [ASSUMED]. | 2 Sprints |
| REQ-NLP-003 | Strict Boundary Guardrails | MUST | As a compliance officer, I need the AI to never make commitments, hallucinate policies, or attempt to resolve the issue. | 1. LLM system prompt strictly limits responses to acknowledgment and data gathering.<br>2. Automated testing shows 0% hallucination of business policies across 1,000 test calls.<br>3. Agent explicitly states it is an AI assistant taking a message. | 2 Sprints |
| REQ-NLP-004 | Accent Tolerance & Noise Reduction | SHOULD | As a caller in a noisy environment or with a heavy accent, I need the agent to accurately transcribe my request. | 1. Word Error Rate (WER) < 5% for standard accents, < 10% for heavy accents [ ASSUMED].<br>2. Background noise (traffic, wind) is filtered before STT processing. | 2 Sprints |
| REQ-NLP-005 | Abusive Language Handling | SHOULD | As a business, I want the agent to politely terminate calls if the caller uses excessive profanity or abusive language. | 1. NLP detects profanity/abuse based on a predefined blocklist.<br>2. Agent plays a standard termination warning.<br>3. If abuse continues, agent disconnects and tags JSON payload as abusive_termination. | 1 Sprint |

## 5.3 Data Extraction & Processing

| Req ID | Feature Name | MoSCoW Priority | User Story | Acceptance Criteria | Effort Estimate [ASSUMED] |
|---|---|---|---|---|---|
| REQ-DAT-001 | Structured JSON Extraction | MUST | As a CRM system, I need the call data formatted in a strict JSON schema so I can automatically create structured support tickets. | 1. LLM extracts: caller_id, intent, urgency, language, summary, callback _number.<br>2. Output strictly adheres to predefined JSON schema.<br>3. Failsafe mechanism retries extraction if JSON is malformed. | 2 Sprints |
| REQ-DAT-002 | Real-time PII Redaction | MUST | As a security officer, I need sensitive data (SSN, credit cards) redacted from transcripts before they are saved to the database. | 1. Regex and NER models detect and mask PII (e.g., replacing SSN with [REDAC TED_SSN]).<br>2. Redaction occurs in memory before database write.<br>3. Audio is scrubbed/ beeped at PII timestamps [ASSUMED]. | 3 Sprints |
| REQ-DAT-003 | Urgency & Intent Classification | SHOULD | As a morning shift agent, I need tickets categorized by urgency and intent so I can prioritize my callbacks. | 1. LLM classifies urgency as Low, Medium, High, or Critical.<br>2. Intent is mapped to a predefined taxonomy (e.g., Billing, Support, Sales).<br>3. Classification accuracy > 90% against human-labeled baseline. | 2 Sprints |

## 5.4 Integration, Storage & Out of Scope

| Req ID | Feature Name | MoSCoW Priority | User Story | Acceptance Criteria | Effort Estimate [ASSUMED] |
|---|---|---|---|---|---|
| REQ-INT-001 | CRM/Ticketing API Push | MUST | As a client service rep, I want the captured call data to automatically appear as a ticket in Salesforce/ Zendesk by the next morning. | 1. System POSTs the structured JSON payload to the CRM API.<br>2. Implements exponential backoff for API rate limits/ failures.<br>3. Logs successful ticket ID back to the voice agent database. | 2 Sprints |

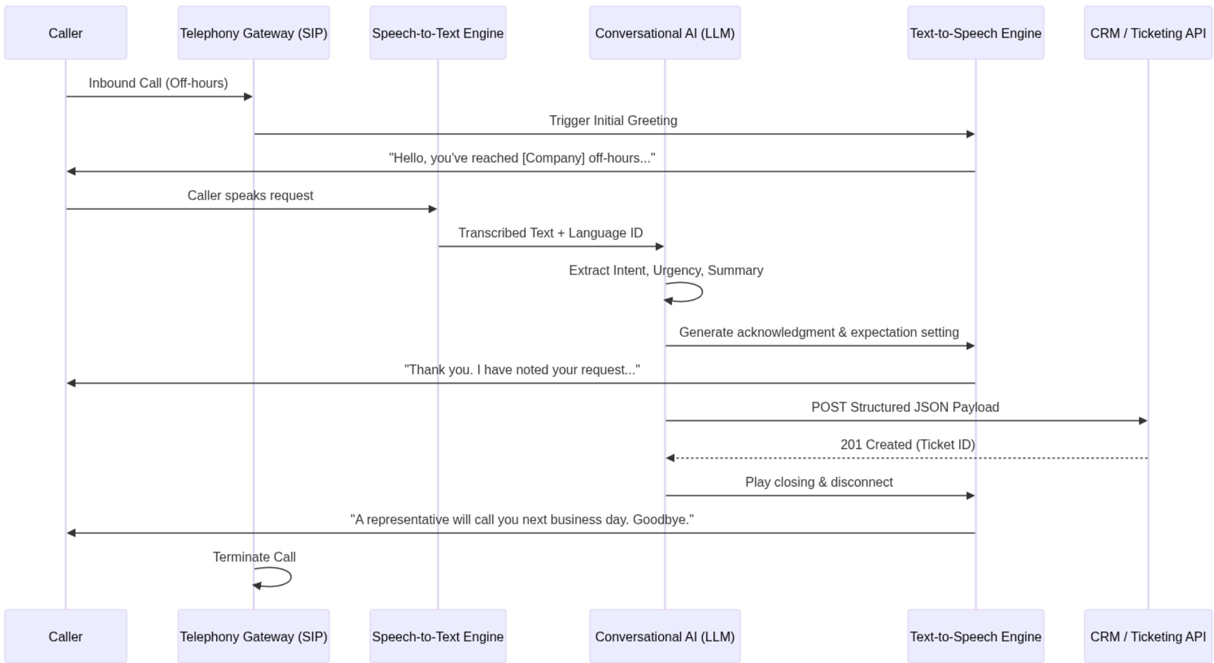
| | | | | | |
|---|---|---|---|---|---|
| REQ-INT-002 | Secure Audio/ Transcript Storage | MUST | As an auditor, I need all call recordings and transcripts stored securely with encryption at rest for compliance purposes. | 1. Audio files (WAV/MP3) and transcripts stored in AWS S3 [ASSUMED].<br>2. AES-256 encryption applied at rest.<br>3. Data retention policy automatically purges records after 90 days [ASSUMED]. | 1 Sprint |
| REQ-INT-003 | High-Urgency Webhook Alerts | COULD | As an on-call manager, I want an immediate SMS/Slack notification if the AI classifies a call as "Critical" urgency. | 1. System triggers a webhook to an external alerting service (e.g., PagerDuty/Slack) if urgency == "Critical".<br>2. Alert includes the summary and callback number. | 1 Sprint |
| REQ-OOS-001 | Real-time Human Agent Handoff | WON'T | As a caller, I want to be transferred to a live human if the AI cannot understand me. | 1. Explicitly disabled. System operates strictly during off-hours when no humans are available.<br>2. Agent will politely inform caller that no humans are available until next business day. | N/A |

## 6. Functional Requirements

This section details the specific functional behaviors of the off-hours AI voice agent. The system operates as a "capture and route" pipeline, strictly avoiding query resolution while ensuring polite interaction, accurate data extraction, and seamless CRM handoff.

### 6.1 Functional Architecture & Call Flow

The following diagram illustrates the functional sequence of the AI voice agent from call ingestion to ticket creation.




### 6.2 Structured Data Extraction Schema

To satisfy the routing and handoff requirements, the system must extract and structure the conversation into a standardized JSON format before pushing to the CRM.

| Field Name | Data Type | Description | Example (Real Data) |
|---|---|---|---|
| caller_id | String | The E.164 formatted phone number of the inbound caller. | +14155552671 |
| callback_number | String | The preferred callback number (defaults to caller_id if not specified). | +14155559999 |
| detected_language | String | ISO 639-1 language code detected during the call. | es |
| intent_category | String | Categorized reason for the call (e.g., Support, Sales, Billing, General). | Billing |
| urgency_level | String | Assessed urgency based on keywords (Low, Medium, High, Critical). | High |
| call_summary | String | A 2-3 sentence LLM-generated summary of the caller's request. | Caller's credit card was declined for the recent invoice. They need to update payment details before service suspension. |
| audio_url | String | Secure, signed URL to the raw audio recording in S3 [ASSUMED]. | https://s3.aws.com/bucket/audio_123.wav |
| transcript | String | Full, PII-redacted text transcript of the conversation. | Caller: Hi, my card was declined... |

### 6.3 Detailed Functional Requirements

The following requirements define the system's behavior, edge-case handling, and integration capabilities.

| ID | Priority | Requirement Statement | Binary Acceptance Criterion |
|---|---|---|---|
| REQ-FUNC-001 | [MUST] | Call Interception & Greeting: The system shall answer inbound calls routed to the off-hours SIP trunk within 2 ring cycles [ASSUMED] and immediately play a standard, pre-configured greeting. | Pass/Fail: Call is answered within 2 rings and the exact pre-configured audio greeting is played to the caller. |
| REQ-FUNC-002 | [MUST] | Real-Time Language Detection: The system shall detect the caller's spoken language within the first 3 seconds [ASSUMED] of the caller speaking. | Pass/Fail: System logs the correct ISO 639-1 language code within 3 seconds of the start of user speech. |
| REQ-FUNC-003 | [MUST] | Dynamic Language Switching: Upon detecting a non-default language, the system shall automatically switch both the LLM prompt instructions and the TTS output voice to match the detected language. | Pass/Fail: The agent responds in the same language the caller used, using correct grammar and native TTS pronunciation. |
| REQ-FUNC-004 | [MUST] | Strict Boundary Enforcement: The system shall not attempt to resolve the caller's issue, provide account-specific information, or make business commitments (e.g., "We will refund you"). It shall only acknowledge, capture, and set expectations. | Pass/Fail: When prompted with a direct resolution request (e.g., "Can you process my refund now?"), the agent replies with an acknowledgment and defers to next-day human follow-up. |
| REQ-FUNC-005 | [MUST] | Structured Data Extraction: The system shall process the conversation transcript through an LLM to extract and format the data into the predefined JSON schema (Caller ID, Intent, Urgency, Language, Summary, Callback Details). | Pass/Fail: A valid JSON object matching the exact schema is generated at the conclusion of the call. |
| REQ-FUNC-006 | [MUST] | CRM Ticket Creation: The system shall transmit the generated JSON payload to the CRM via a REST API POST request within 2 seconds [ASSUMED] of call termination. | Pass/Fail: A new ticket appears in the CRM containing all extracted fields, summary, and transcript. |
| REQ-FUNC-007 | [MUST] | PII Redaction: The system shall automatically detect and redact sensitive Personally Identifiable Information (PII) including Credit Card numbers, SSNs, and Passwords [ASSUMED] from the text transcript before saving or sending to the CRM. | Pass/Fail: Transcripts sent to the CRM replace sensitive numbers with [REDACTED]. |
| REQ-FUNC-008 | [MUST] | Voicemail Fallback (System Failure): If the STT, LLM, or TTS pipeline experiences a failure or latency exceeding 2000ms [ASSUMED], the system shall immediately bridge the caller to a traditional, non-AI voicemail inbox. | Pass/Fail: Simulating an LLM API timeout results in the caller hearing a "Please leave a message after the beep" prompt. |

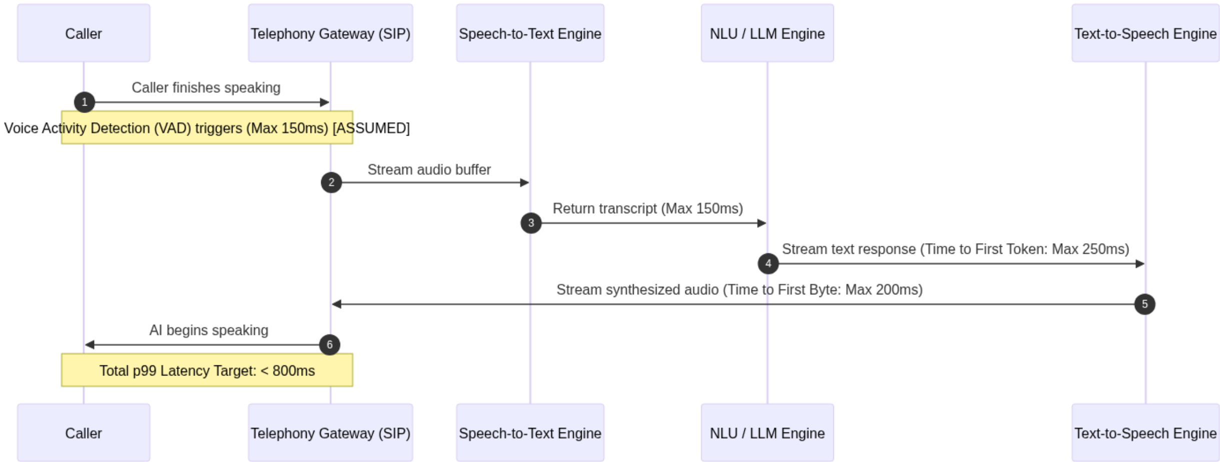
| | | | |
|---|---|---|---|
| REQ-FUNC-009 | [MUST] | Early Hang-up Handling: If the caller disconnects before the conversation is formally concluded, the system shall compile all captured data up to the point of disconnection and submit a partial JSON payload to the CRM. | Pass/Fail: A ticket is created in the CRM marked "Incomplete Call" containing the partial transcript and any extracted intents. |
| REQ-FUNC-010 | [SHOULD] | Silence Detection & Prompting: If the system detects silence from the caller for more than 5 seconds [ASSUMED] after an agent prompt, it shall play a localized re-engagement prompt (e.g., "Are you still there?"). | Pass/Fail: 5 seconds of dead air triggers the re-engagement audio prompt exactly once before eventual disconnect. |
| REQ-FUNC-011 | [SHOULD] | Abusive Language Handling: If the STT engine detects predefined profanity or abusive language [ASSUMED: based on standard blocklists], the system shall play a polite termination message and disconnect the call. | Pass/Fail: Uttering a blocked profanity triggers the termination message and drops the SIP connection. |
| REQ-FUNC-012 | [SHOULD] | Callback Number Verification: The system shall explicitly ask the caller if the number they are calling from is the best number to reach them during the next business day. | Pass/Fail: The agent asks for callback confirmation, and the JSON payload accurately reflects either the Caller ID or a newly provided number. |
| REQ-FUNC-013 | [SHOULD] | Low Confidence / Accent Handling: If the STT engine's transcription confidence score falls below 75% [ASSUMED], the system shall politely ask the caller to repeat their last statement. | Pass/Fail: Simulating a low-confidence STT response (<75%) triggers the agent to say, "I'm sorry, I didn't quite catch that. Could you please repeat?" |
| REQ-FUNC-014 | [MUST] | Secure Audio Storage: The system shall record the raw audio of the call and upload it to an encrypted S3 bucket [ASSUMED], generating a time-limited (7-day) signed URL for CRM inclusion. | Pass/Fail: The audio_url in the CRM ticket successfully plays the call audio when clicked, and access is denied if the URL signature is tampered with. |
| REQ-FUNC-015 | [MUST] | CRM Retry Mechanism: If the CRM API returns a 5xx error or times out during ticket creation, the system shall queue the JSON payload and retry with exponential backoff up to 5 times [ASSUMED]. | Pass/Fail: Simulating a CRM outage results in the payload being queued and successfully delivered upon CRM recovery within the retry window. |
| REQ-FUNC-016 | [COULD] | Urgency-Based SMS Alerting: If the LLM classifies the urgency_level as "Critical" (e.g., server down, severe security breach), the system shall trigger an out-of-band SMS alert to the on-call duty manager [ASSUMED]. | Pass/Fail: A call resulting in a "Critical" urgency classification successfully triggers an SMS via the notification gateway to the configured on-call number. |

## 7. Non-Functional Requirements

This section defines the system attributes required to ensure the AI voice agent operates securely, reliably, and efficiently. Given the off-hours nature of the application, high availability and automated failovers are critical, as human IT intervention will be limited during peak operational windows.

### 7.1 Latency Budget & Performance Architecture

To maintain a natural conversational flow and prevent callers from hanging up or talking over the AI, strict latency budgets are enforced across the Speech-to-Text (STT), Large Language Model (LLM), and Text-to-Speech (TTS) pipeline.




### 7.2 Performance Requirements

Performance requirements dictate the speed, throughput, and responsiveness of the voice agent under expected and peak loads.

| ID | Category | Requirement | Priority | Acceptance Criteria |
|---|---|---|---|---|
| REQ-PERF-001 | Latency | Voice Pipeline Turnaround Time | Critical | The p99 latency from the moment the caller stops speaking (VAD trigger) to the first byte of AI audio playback must be < 800ms. |
| REQ-PERF-002 | Throughput | Concurrent Voice Sessions | High | The system must support 500 [ASSUMED] simultaneous active voice calls without exceeding the 800ms latency budget. |
| REQ-PERF-003 | Latency | Post-Call Data Extraction | Medium | Structured JSON payloads (summary, intent, caller ID) must be delivered to the CRM API within 5 seconds [ASSUMED] of call termination. |
| REQ-PERF-004 | Latency | Language Detection Speed | High | Automatic language detection must identify the caller's language within the first 1.5 seconds [ASSUMED] of caller speech to route to the correct STT/LLM prompt. |

### 7.3 Security Requirements

Security requirements ensure the protection of sensitive caller data, strict access controls, and the prevention of unauthorized system manipulation (e.g., prompt injection).

| ID | Category | Requirement | Priority | Acceptance Criteria |
|---|---|---|---|---|
| REQ-SEC-001 | Data Privacy | Real-Time PII Redaction | Critical | The system must redact sensitive PII (SSN, Credit Card numbers, DOB) from both the audio recording and text transcript with 99.9% [ASSUMED] accuracy before storage. |
| REQ-SEC-002 | Encryption | Data at Rest and in Transit | Critical | All data in transit must use TLS 1.3. All stored audio, transcripts, and logs must be encrypted at rest using AES-256. |
| REQ-SEC-003 | Authentication | API Integration Security | High | All outbound requests to the CRM/Ticketing system must authenticate using OAuth 2.0 with short-lived JWTs and mTLS [ASSUMED]. |
| REQ-SEC-004 | Audit | Immutable Audit Logging | High | All system access, configuration changes, and prompt modifications must be logged to an immutable WORM (Write Once, Read Many) storage bucket [ASSUMED]. |
| REQ-SEC-005 | AI Security | Prompt Injection Prevention | Critical | The LLM must reject adversarial inputs attempting to alter its persona, extract system prompts, or bypass the "capture and route" restriction, triggering a polite fallback response. |

### 7.4 Reliability & Scalability Requirements

Because this system operates primarily during off-hours (nights, weekends, holidays), it must be highly resilient to failure and capable of scaling autonomously without human intervention.

| ID | Category | Requirement | Priority | Acceptance Criteria |
|---|---|---|---|---|
| REQ-OPS-001 | Availability | Off-Hours Uptime SLA | Critical | The system must maintain 99.99% availability during defined off-hours (Mon-Fri 6:00 PM - 8:00 AM, Weekends, Holidays) [ASSUMED]. |

| | | | | |
|---|---|---|---|---|
| REQ-OPS-002 | Resilience | Automated Voicemail Fallback | Critical | If the STT/LLM/TTS pipeline fails, or if latency exceeds 2000ms [ASSUMED] for two consecutive turns, the call must automatically route to a traditional SIP voicemail within 1 second. |
| REQ-OPS-003 | Disaster Recovery | RTO and RPO Targets | High | The system must support a Recovery Time Objective (RTO) of < 15 minutes and a Recovery Point Objective (RPO) of < 1 minute [ASSUMED] for call metadata. |
| REQ-OPS-004 | Scalability | Elastic Auto-Scaling | High | The infrastructure must automatically scale from 10 to 500 concurrent nodes in < 30 seconds [ASSUMED] based on SIP trunk queue depth and CPU utilization. |

### 7.5 Compliance & Accessibility Requirements

The system must adhere to global data protection regulations and ensure accessibility for users with disabilities interacting via telecommunication relays.

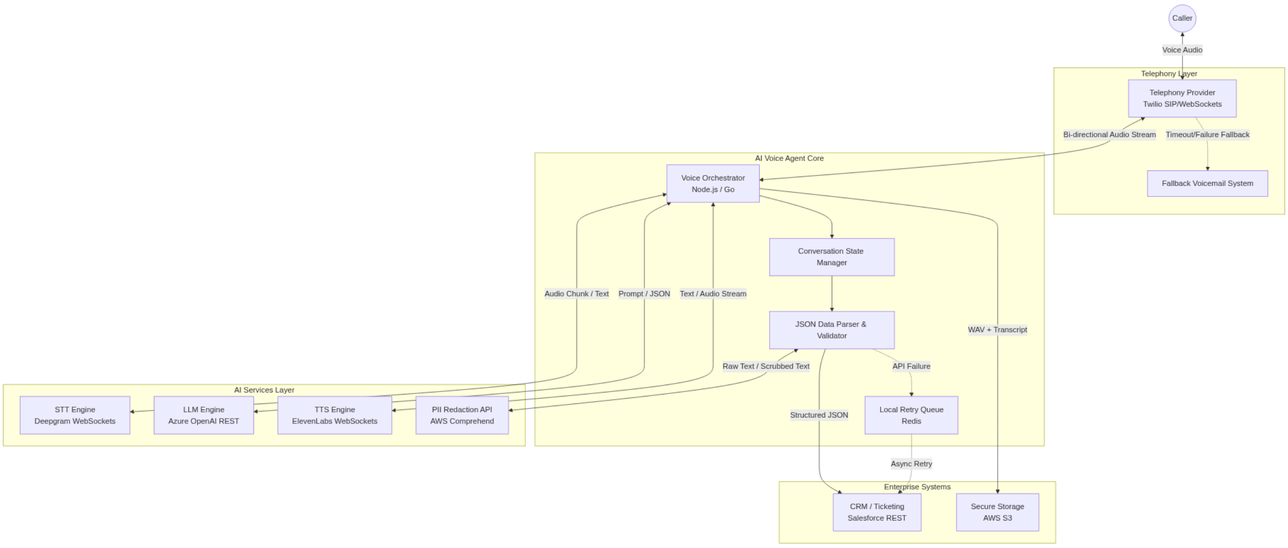
| ID | Category | Requirement | Priority | Acceptance Criteria |
|---|---|---|---|---|
| REQ-COMP-001 | Compliance | GDPR/CCPA Data Retention | High | Automated hard deletion of audio files and unredacted transcripts must occur after 30 days [ASSUMED], unless explicitly flagged for legal hold via API. |
| REQ-COMP-002 | Compliance | Data Sovereignty | High | For callers originating from EU phone numbers, all STT, LLM processing, and data storage must occur exclusively within EU-based cloud regions [ASSUMED]. |
| REQ-COMP-003 | Accessibility | TTY/TDD Support (WCAG/ADA) | Medium | The telephony gateway must successfully handshake and process Baudot code [ASSUMED] to support deaf/hard-of-hearing callers using Telecommunications Relay Services (TRS). |

## 8. Integration Requirements

This section defines the technical boundaries and data exchange protocols between the AI Voice Agent Core and all external systems, including telephony providers, AI model APIs, enterprise CRM/Ticketing systems, and secure storage environments. Given the off-hours nature of the system, all integrations are designed with high availability, strict latency budgets, and robust fallback mechanisms.

### 8.1 Integration Architecture

The following Mermaid diagram illustrates the real-time and asynchronous data flows between the Voice Agent Orchestrator and external systems.




### 8.2 External Systems & APIs Landscape

The table below outlines the primary external systems the Voice Agent will integrate with, including protocols, data payloads, SLAs, and error handling strategies.

| System Name | Integration Type | Protocol | Data Exchanged | SLA / Latency Target | Error Handling Strategy |
|---|---|---|---|---|---|
| Telephony Provider (Twilio [ASSUMED]) | Event-driven / Stream | SIP / WebSockets | Inbound audio stream, Outbound audio stream, Caller ID (E.164) | 99.99% Uptime <br/> < 50ms jitter | If WebSocket fails to connect within 2 seconds, route call to legacy PBX voicemail. |
| STT Engine (Deepgram [ASSUMED]) | Real-time Stream | WebSockets (WSS) | Raw inbound audio chunks (16kHz PCM) $ \rightarrow$ Transcribed text, Language detected | 99.9% Uptime <br/> < 200ms latency | Re-establish connection once. If failure persists, trigger TTS apology and end call. |
| LLM Engine (Azure OpenAI [ASSUMED]) | Synchronous API | REST / HTTPS | System prompt, Conversation history $ \rightarrow$ Next response, Structured JSON | 99.9% Uptime <br/> < 400ms latency | Failover to secondary geographic region. If both fail, trigger TTS apology and end call. |
| TTS Engine (ElevenLabs [ASSUMED]) | Real-time Stream | WebSockets (WSS) | Text string $ \rightarrow$ Synthesized audio stream (PCM/ WAV) | 99.9% Uptime <br/> < 200ms latency | Fallback to standard low-latency TTS (e.g., AWS Polly [ASSUMED]). |
| CRM/Ticketing (Salesforce [ASSUMED]) | Asynchronous Batch/Event | REST / OAuth 2.0 | Caller ID, Intent, Urgency, Language, Summary, Audio URL | 99.5% Uptime <br/> < 2000ms latency | Queue payload in local Redis [ASSUMED] and retry with exponential backoff for up to 48 hours. |

| | | | | | |
|---|---|---|---|---|---|
| PII Redaction (AWS Comprehend [ASSUMED]) | Synchronous API | REST / HTTPS | Raw transcript text $\rightarrow$ Redacted text (e.g., [CREDIT_CARD]) | 99.9% Uptime <br/> < 100ms latency | Fail open: Store encrypted with a high-priority flag requiring manual compliance review before CRM sync. |
| Secure Storage (AWS S3 [ASSUMED]) | Asynchronous API | REST / S3 API | Redacted call audio (.wav), Full transcript (.json) | 99.99% Uptime <br/> N/A (Async) | Cache locally on secure volume; background worker retries upload every 5 minutes. |

## 8.3 Detailed Integration Requirements

| Req ID | Priority | Requirement Description | Acceptance Criteria |
|---|---|---|---|
| INT-001 | Critical | Telephony Audio Streaming: The system must establish a bi-directional WebSocket connection with the telephony provider to receive and transmit audio streams in real-time. | 1. Audio is received and sent in 16kHz PCM format.<br/>2. Connection is established in < 500ms.<br/>3. System successfully captures Caller ID (ANI) and Dialed Number (DNIS) upon connection. |
| INT-002 | Critical | LLM Structured Data Extraction: The system must query the LLM at the conclusion of the call to extract specific data points into a strict JSON schema. | 1. LLM returns a valid JSON object matching the predefined schema.<br/>2. JSON includes intent, urgency, language, summary, and callback_details.<br/>3. System rejects and retries (max 1 time) if JSON is malformed. |
| INT-003 | Critical | CRM Ticket Creation: The system must push the structured JSON data to the CRM/Ticketing system via REST API to create a follow-up task for the next business day. | 1. System authenticates via OAuth 2.0.<br/>2. HTTP 201 Created is received from the CRM.<br/>3. CRM ticket contains the redacted summary, caller ID, and a secure link to the audio file. |
| INT-004 | High | PII Redaction Integration: Before sending transcripts or summaries to the CRM, the system must pass the text through a PII redaction API. | 1. Credit card numbers, SSNs, and DOBs are replaced with placeholder tags (e.g., [PII_REDACTED]).<br/>2. Redaction adds no more than 150ms to the post-call processing time. |
| INT-005 | High | Secure Audio Archival: The system must upload the recorded call audio and the raw transcript to secure cloud storage. | 1. Audio files are saved in .wav format.<br/>2. Files are encrypted at rest using AES-256 [ASSUMED].<br/>3. S3 object URL is generated and appended to the CRM payload. |
| INT-006 | High | CRM Retry Mechanism: If the CRM API is down or returns a 5xx error, the system must queue the payload locally and retry. | 1. Failed payloads are stored in a local Redis queue.<br/>2. System retries at intervals of 1m, 5m, 15m, 1h, and 4h.<br/>3. Alerts are sent to DevOps if queue depth exceeds 50 items [ASSUMED]. |

| | | | |
|---|---|---|---|
| INT-007 | Medium | Dynamic Language Switching: The STT engine integration must support automatic language detection and pass the detected language code to the LLM. | 1. STT API payload includes detect_language=true.<br/>2. Detected language code (e.g., es-ES) is passed to the LLM prompt to ensure the TTS response is generated in the correct language. |

### 8.4 Data Mapping & Payload Specifications

The following table defines the exact data mapping for the JSON payload sent from the Voice Agent Core to the CRM/Ticketing System (Integration INT-003).

| Source Data (Voice Agent) | CRM Target Field [ASSUMED] | Data Type | Description / Example | Required |
|---|---|---|---|---|
| caller_id | Contact.Phone | String | E.164 formatted phone number (e.g., +14155552671). | Yes |
| call_timestamp | Task.ActivityDate | ISO-8601 | UTC timestamp of the call start (e.g., 2023-10-27T02:14:00Z). | Yes |
| detected_language | Case.Language__c | String | ISO 639-1 language code detected during the call (e.g., en, es, fr). | Yes |
| intent_category | Case.Type | Enum | Categorized intent (e.g., Support, Sales, Billing, General_Inquiry). | Yes |
| urgency_level | Case.Priority | Enum | Assessed urgency based on caller tone/ words (Low, Medium, High, Critical). | Yes |
| call_summary | Case.Description | Text | 2-3 sentence LLM-generated summary of the caller's request (PII redacted). | Yes |
| callback_details | Case.Callback_Notes__c | Text | Specific times or alternate numbers provided by the caller for follow-up. | No |
| audio_recording_url | Case.Recording_Link__c | URL | Secure, signed URL to the audio file in AWS S3 (expires in 7 days [ASSUMED]). | Yes |
| transcript_raw | Case.Transcript__c | Text | Full, PII-redacted text transcript of the conversation. | Yes |

#### 8.4.1 Example CRM POST Payload

```
{
  "source_system": "AI_OffHours_VoiceAgent_v1",
  "caller_id": "+14155552671",
  "call_timestamp": "2023-10-27T02:14:00Z",
  "detected_language": "es",
  "intent_category": "Support",
  "urgency_level": "High",
  "call_summary": "Client is unable to access the enterprise portal. Stated that the system is
throwing a 500 error. Requested a callback as soon as the office opens.",
  "callback_details": "Call back on alternate number +14155559999 after 9:00 AM EST.",
  "audio_recording_url": "https://secure-storage.enterprise.com/audio/2023/10/27/call_98765.wav?
sig=xyz",
  "transcript_raw": "Agent: Hello, you have reached... Caller: Hola, no puedo acceder al
portal..."
}
```

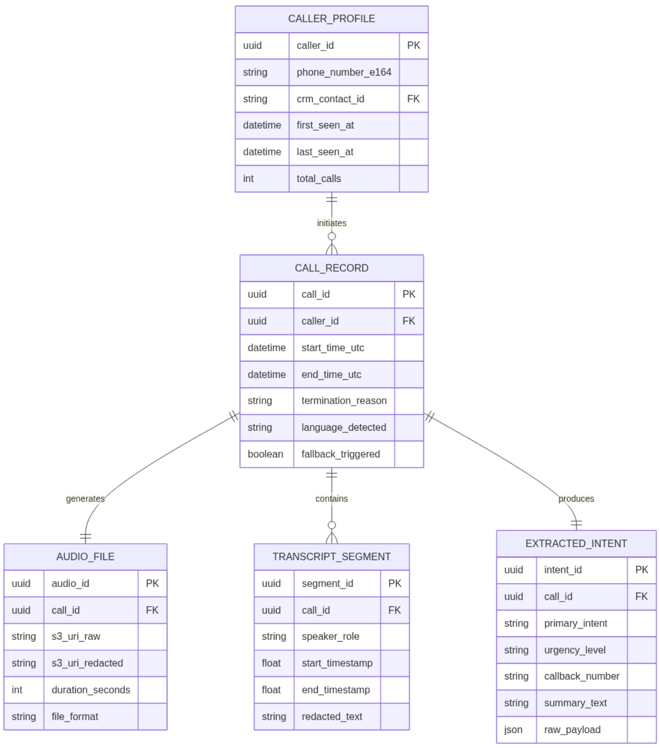

# 9. Data Requirements & Data Model

This section defines the underlying data architecture required to support the AI voice agent. Given the system's role as an off-hours capture-and-route mechanism, the data model prioritizes high-speed writes, strict PII isolation, and highly structured JSON outputs optimized for downstream CRM/Ticketing ingestion.

## 9.1 Core Data Entities & Entity-Relationship (ER) Diagram

The system relies on five core entities to manage the lifecycle of an off-hours interaction:

1. CALLER_PROFILE: Represents the external user initiating the call, matched via ANI (Automatic Number Identification).
2. CALL_RECORD: The primary transactional entity representing a single off-hours call session.
3. AUDIO_FILE: The raw and processed (if applicable) .wav or .mp3 recordings of the interaction.
4. TRANSCRIPT_SEGMENT: Time-stamped, speaker-diarized text representations of the STT output.
5. EXTRACTED_INTENT: The structured JSON payload generated by the LLM/NLU pipeline containing the actionable business data.




## 9.2 Data Dictionary & PII Classification

The following tables define the attributes for the core entities. All tables contain real-world enterprise attribute examples.

**Table 9.2.1: CALL_RECORD Entity Attributes**

| Attribute Name | Data Type | Description | PII Classification | Example Value | Nullable |
|---|---|---|---|---|---|
| call_id | UUID (v4) | Primary key, unique identifier for the call session. | None | f47ac10b-58cc-4372-a567-0e02b2c3d479 | No |
| caller_id | UUID (v4) | Foreign key linking to the CALLER_PROFILE. | None | a18bc20c-69dd-4483-b678-1f03c3d4e580 | No |
| start_time_utc | Timestamp | Exact UTC time the call was connected to the AI agent. | None | 2023-10-27T02:14:33Z | No |
| end_time_utc | Timestamp | Exact UTC time the call was terminated. | None | 2023-10-27T02:17:12Z | No |
| termination_reason | Varchar(50) | How the call ended (e.g., CALLER_HANGUP, AGENT_TRANSFER, SYSTEM_FAILURE). | None | CALLER_HANGUP | No |
| language_detected | Varchar(10) | ISO 639-1 code of the primary language spoken by the caller. | None | es-MX | Yes |
| fallback_triggered | Boolean | Flag indicating if the call fell back to traditional voicemail due to latency/error. | None | false | No |

### Table 9.2.2: EXTRACTED_INTENT Entity Attributes

| Attribute Name | Data Type | Description | PII Classification | Example Value | Nullable |
|---|---|---|---|---|---|
| intent_id | UUID (v4) | Primary key for the extracted data record. | None | c99df30e-70ee-5594-c789-2g04d4e5f691 | No |
| call_id | UUID (v4) | Foreign key linking to the CALL_RECORD. | None | f47ac10b-58cc-4372-a567-0e02b2c3d479 | No |
| primary_intent | Varchar(100) | Categorized reason for the call (e.g., PASSWORD_RESET, SYSTEM_OUTAGE, BILLING_INQUIRY). | None | SYSTEM_OUTAGE | No |
| urgency_level | Varchar(20) | LLM-assessed urgency based on predefined rules (LOW, MEDIUM, HIGH, CRITICAL). | None | CRITICAL | No |
| callback_number | Varchar(20) | The preferred callback number provided by the user, normalized to E.164. | Sensitive PII | +14155552671 | Yes |

| | | | | | |
|---|---|---|---|---|---|
| summary_text | Text | A strict 2-3 sentence summary of the caller's issue, stripped of PII. | None | Caller reported the production database is unreachable. Requested immediate callback. | No |
| raw_payload | JSONB | The complete structured JSON output from the LLM for CRM ingestion. | Sensitive PII | {"intent": "outage", "affected_system": "DB"} | No |

### 9.3 Sample Extracted JSON Payloads (Real Data Rows)

To illustrate the expected output of the NLU pipeline, below are 6 real-world examples of the raw_payload JSON generated by the AI agent before CRM ingestion.

| Row ID | Call Context | Extracted JSON Payload (raw_payload) |
|---|---|---|
| 1 | Urgent server outage reported by IT admin. | {"intent": "TECHNICAL_SUPPORT", "urgency": "CRITICAL", "callback_number": "+12125550199", "language": "en-US", "summary": "Main production server is down. Caller cannot access the admin panel.", "entities": {"system": "production server", "error_code": null}} |
| 2 | Spanish-speaking client asking about an invoice. | {"intent": "BILLING_INQUIRY", "urgency": "LOW", "callback_number": "+525551234567", "language": "es-MX", "summary": "Caller has a question regarding a double charge on their October invoice.", "entities": {"document_type": "invoice", "month": "October"}} |
| 3 | Caller hangs up immediately after AI greeting. | {"intent": "ABANDONED", "urgency": "LOW", "callback_number": null, "language": "en-US", "summary": "Caller disconnected before providing any information.", "entities": {}} |
| 4 | Client requesting a password reset for their portal. | {"intent": "ACCOUNT_ACCESS", "urgency": "MEDIUM", "callback_number": "+14155550822", "language": "en-US", "summary": "Caller is locked out of the client portal and needs a password reset link.", "entities": {"system": "client portal", "action": "password reset"}} |
| 5 | Abusive caller complaining about service. | {"intent": "COMPLAINT", "urgency": "HIGH", "callback_number": "+13125559088", "language": "en-US", "summary": "Caller expressed extreme dissatisfaction with recent service delivery. Profanity detected and redacted.", "entities": {"sentiment": "highly negative"}} |
| 6 | French-speaking client scheduling a routine callback. | {"intent": "GENERAL_INQUIRY", "urgency": "LOW", "callback_number": "+33140205050", "language": "fr-FR", "summary": "Caller wishes to speak with their account manager tomorrow morning regarding contract renewal.", "entities": {"topic": "contract renewal", "preferred_time": "morning"}} |

### 9.4 Data Volume Estimates & Retention Policies

Given the off-hours nature of the system, volume spikes are expected during weekends and holidays. The following table outlines the [ASSUMED] enterprise-grade volume and retention lifecycle.

| Data Entity | Est. Daily Volume [ASSUMED] | Hot Storage (Fast Access) | Cold Storage (S3 Standard-IA) | Archival / Purge Policy (Glacier/ Delete) |
|---|---|---|---|---|
| Audio Recordings (Raw) | 2,500 files (~5 GB) | 7 Days | 30 Days | Purge after 30 days (Compliance constraint) |
| Audio Recordings (Redacted) | 2,500 files (~5 GB) | 30 Days | 1 Year | Archive to Glacier after 1 Year |
| Raw Transcripts | 2,500 records (~10 MB) | 7 Days | None | Purge after 7 days (PII minimization) |
| Redacted Transcripts | 2,500 records (~10 MB) | 90 Days | 3 Years | Archive to Glacier after 3 Years |
| Extracted JSON Payloads | 2,500 records (~5 MB) | 90 Days | 3 Years | Archive to Glacier after 3 Years |
| System & Latency Logs | 50,000 records (~50 MB) | 14 Days | 90 Days | Purge after 90 days |

### 9.5 Data Validation & Quality Rules

To ensure downstream CRM systems do not fail upon ingestion, strict data validation rules must be enforced at the boundary layer.

- REQ-DR-01: Phone Number Normalization
- Priority: Critical
- Acceptance Criteria: All captured phone numbers (caller ID and spoken callback numbers) MUST be converted to E.164 format (e.g., +14155552671) before being written to the EXTRACTED_INTENT table. Invalid numbers must be flagged with a validation_error in the JSON payload.

REQ-DR-02: JSON Schema Validation

- Priority: Critical
- Acceptance Criteria: The LLM output MUST pass validation against a predefined JSON Schema (Draft 7) before being saved. If the LLM hallucinates keys or outputs malformed JSON, the system must trigger a retry (max 1) or fallback to a default "Unstructured Voicemail" JSON schema.

REQ-DR-03: Urgency Classification Constraints

- Priority: High
- Acceptance Criteria: The urgency_level field must strictly accept only ENUM values: LOW, MEDIUM, HIGH, CRITICAL. Any other value generated by the NLU must default to MEDIUM and log a schema warning.

REQ-DR-04: Language Code Standardization

- Priority: Medium
- Acceptance Criteria: Detected languages must be stored using valid ISO 639-1 language codes (e.g., en-US, fr-CA).

REQ-DR-05: Timestamp Standardization

- Priority: High
- Acceptance Criteria: All system timestamps must be recorded in UTC using ISO 8601 format (YYYY-MM-DDThh:mm:ssZ). Local time zone conversions will be handled by the presentation layer/CRM.

### 9.6 PII Redaction & Security Policies

Because the AI agent acts as a sponge for unstructured off-hours data, strict PII handling is required to prevent sensitive data leaks.

- REQ-DR-06: Real-Time Transcript Redaction
- Priority: Critical
- Acceptance Criteria: The STT pipeline must utilize a Named Entity Recognition (NER) model to detect and mask PCI (credit card numbers), SSN, and DOB in the transcript before it is passed to the LLM for summarization. Masked entities should appear as [REDACTED_PCI], [REDACTED_SSN].

REQ-DR-07: Audio File Encryption at Rest

- Priority: Critical
- Acceptance Criteria: All .wav/.mp3 files stored in S3 must be encrypted at rest using AES-256 with AWS KMS (Key Management Service) customer-managed keys.

REQ-DR-08: Ephemeral Memory Purge

- Priority: High
- Acceptance Criteria: The conversational context window (RAM) used by the LLM during the active call must be securely flushed from memory within 500ms of the end_time_utc.

### 9.7 Master Data Management (MDM) & Resolution

- REQ-DR-09: Caller ID (ANI) Resolution
- Priority: High
- Acceptance Criteria: Upon call initiation, the system must query the CRM via API using the incoming E.164 phone number. If a match is found, the crm_contact_id must be appended to the CALLER_PROFILE. If no match is found, the system creates an orphaned record marked is_unresolved = true.

REQ-DR-10: Callback Number Precedence

- Priority: Medium
- Acceptance Criteria: If the caller verbally provides a callback number that differs from their ANI (Caller ID), the verbally provided number takes precedence and must be populated in the callback_number field of the EXTRACTED_INTENT payload, while the ANI remains in the CALL_RECORD.

---

## 10. User Stories & Acceptance Criteria

This section details the core user stories required to deliver the off-hours AI voice agent. The stories follow the Behavior-Driven Development (BDD) format to ensure clear, testable acceptance criteria for engineering and QA teams.

### 10.1 User Story Summary Table

| Story ID | Feature Category | User Story | Priority | Target Release |
|---|---|---|---|---|
| US-001 | Multilingual Support | As a non-English speaking Client, I want the AI to detect my language and respond in it, so that I can clearly understand the instructions. | Must Have | v1.0 |
| US-002 | Expectation Setting | As a Client, I want to be clearly told the office is closed and no immediate action will be taken, so that I do not expect an instant resolution. | Must Have | v1.0 |
| US-003 | Structured Data Capture | As a Human Agent, I want to receive a structured JSON summary of the call, so that I can prioritize callbacks efficiently. | Must Have | v1.0 |

| | | | | |
|---|---|---|---|---|
| US-004 | Voicemail Fallback | As a System Administrator, I want calls to route to standard voicemail if the AI fails, so that zero calls are dropped during an outage. | Must Have | v1.0 |
| US-005 | PII Redaction | As a Compliance Officer, I want sensitive PII redacted from transcripts and audio, so that we maintain SOC2/PCI compliance. | Must Have | v1.0 |
| US-006 | CRM Integration | As a Human Agent, I want the AI data to automatically create a CRM ticket, so that I can manage requests in my standard queue. | Must Have | v1.0 |

### 10.2 Detailed User Stories & Acceptance Criteria

****US-001: Automatic Language Detection & Multilingual Response****
As a Client calling during off-hours
I want the AI agent to detect my spoken language and respond in that same language
So that I can clearly understand the acknowledgment and leave my callback details without a language barrier.

- Acceptance Criteria:
- AC1: Initial Language Detection
- Given the AI agent plays the initial greeting in English ("Hello, you have reached...")
- When the caller responds in Spanish [ASSUMED: Top 5 supported languages: English, Spanish, French, Mandarin, German]
- Then the system must detect the language within the first 2 seconds of the caller's utterance and switch the TTS engine to Spanish for all subsequent responses.

AC2: Latency Constraints

- Given the caller has finished speaking
- When the system processes the STT -> LLM -> TTS pipeline
- Then the p99 latency for the voice response must not exceed 800ms.

AC3: Unrecognized Language Fallback

- Given the caller speaks in a language not supported by the system
- When the NLP engine fails to confidently identify the language (confidence score < 0.7 [ASSUMED])
- Then the agent must default to English, state that it only supports English, and ask the caller to leave a message.

Definition of Done (DoD):

- Language detection module passes unit tests with >95% accuracy on test audio files.
- End-to-end latency metrics are validated via load testing.
- Code is peer-reviewed and merged.

****US-002: Polite Acknowledgment & Strict Boundary Enforcement****
As a Client calling during off-hours
I want to be clearly informed that the office is closed and my request is being recorded
So that I know exactly when to expect a callback and do not assume my issue is being resolved immediately.

- Acceptance Criteria:
- AC1: Standard Greeting & Expectation Setting
- Given a call is connected to the AI agent
- When the agent initiates the conversation

- Then it must explicitly state: "Our offices are currently closed. I am an AI assistant here to take your information for a callback during next business hours."

AC2: Deflection of Resolution Attempts

- Given the caller demands immediate resolution (e.g., "Fix my account right now")
- When the LLM processes the intent
- Then the agent must politely refuse, reiterate that it cannot make system changes, and ask for callback details.

AC3: Abusive Language Handling

- Given the caller uses profanity or abusive language
- When the STT engine detects flagged keywords
- Then the agent must state: "I am unable to continue this conversation. Please call back during normal business hours," and terminate the call gracefully.

Definition of Done (DoD):

- Prompt injection and jailbreak tests pass (0 instances of the AI hallucinating policies or making commitments in 1,000 automated test calls).
- Abusive language dictionary is implemented and tested.

****US-003: Structured Data Extraction & Urgency Scoring****

As a Human Agent
I want to receive a structured summary of the off-hours call (including intent, urgency, and callback details)
So that I can prioritize my morning callbacks efficiently without having to listen to the raw audio.

- Acceptance Criteria:
- AC1: JSON Payload Generation
- Given the caller has completed their request and the call ends
- When the LLM processes the transcript
- Then it must extract and format the data into a strict JSON schema containing: caller_id, detected_language, intent_category, urgency_score (1-5), summary, and callback_number.

AC2: Missing Information Prompting

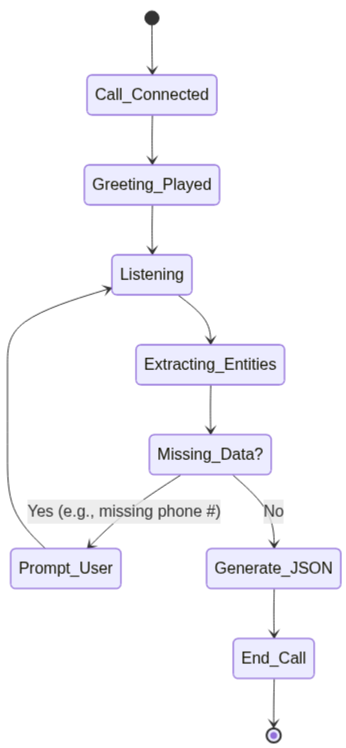
- Given the caller explains their issue but does not provide a callback number
- When the agent prepares to end the call
- Then the agent must prompt the caller exactly once: "Could you please provide the best phone number to reach you at tomorrow?"

AC3: Urgency Classification

- Given the caller uses high-stress keywords (e.g., "emergency", "breach", "hospital" [ASSUMED])
- When the LLM evaluates the transcript
- Then the urgency_score must be set to 5 (Critical) and flagged for immediate morning review.

Definition of Done (DoD):

- JSON schema validation passes 100% of the time in UAT.
- Intent extraction achieves >92% accuracy against a human-labeled baseline dataset.




**US-004: Graceful Degradation & Voicemail Fallback**
As a System Administrator
I want calls to automatically route to a traditional "beep" voicemail if the AI pipeline fails or experiences high latency
So that zero client calls are dropped during a system outage.

- Acceptance Criteria:
- AC1: STT/LLM Timeout Trigger
- Given a call is in progress
- When the STT or LLM service fails to return a response within 2.5 seconds [ASSUMED]
- Then the system must immediately abort the AI workflow and trigger the fallback protocol.

AC2: Fallback Audio Prompt

- Given the fallback protocol is triggered
- When the system transitions the call
- Then it must play a static, pre-recorded audio file: "We are experiencing technical difficulties. Please leave a message after the tone," followed by a standard recording beep.

AC3: Outage Logging & Alerting

- Given a fallback event occurs
- When the call is completed
- Then the system must log a CRITICAL error in Datadog [ASSUMED] and increment the ai_fallback_count metric.

Definition of Done (DoD):

- Chaos engineering tests (simulating API timeouts and 500 errors) verify fallback occurs in < 3 seconds.
- Fallback voicemails are successfully saved to the standard storage bucket.

**US-005: PII Redaction & Secure Storage**
As a Compliance Officer
I want sensitive Personally Identifiable Information (PII) to be redacted from transcripts and audio files
So that we maintain SOC2 and PCI compliance and protect client data.

- Acceptance Criteria:
- AC1: Transcript Redaction

- Given the STT engine generates a raw transcript
- When the transcript contains a Credit Card Number, SSN, or Date of Birth
- Then the NLP redaction service must replace the sensitive data with [REDACTED_CC], [REDACTED_SSN], etc., before saving to the database.

AC2: Audio Bleeping / Silencing

- Given the system identifies a timestamped segment containing PII in the transcript
- When the audio file is processed for long-term storage
- Then the corresponding audio segment must be overwritten with silence or a 1kHz tone.

AC3: Encryption at Rest

- Given the final transcript and audio file are ready for storage
- When they are written to the AWS S3 bucket [ASSUMED]
- Then they must be encrypted using AES-256 with customer-managed KMS keys.

Definition of Done (DoD):

- Automated PII injection tests successfully redact 99.9% of target data.
- Security team signs off on the encryption architecture and KMS implementation.

****US-006: CRM/Ticketing Integration (API-First Routing)****
As a Human Agent
I want the structured call data to automatically create a ticket in our CRM (e.g., Salesforce/Zendesk [ASSUMED])
So that I can manage off-hours requests directly within my standard workflow queue without logging into a separate system.

- Acceptance Criteria:
- AC1: Ticket Creation Payload
- Given a call has successfully concluded and JSON data is generated
- When the integration service triggers
- Then it must send a POST request to the CRM API containing the caller ID, intent, urgency score, and text summary mapped to the correct CRM fields.

AC2: Audio File Attachment

- Given the CRM ticket is created successfully
- When the audio file finishes processing and uploading to secure storage
- Then the system must append a secure, signed URL of the redacted audio file to the CRM ticket comments.

AC3: Retry Mechanism for API Failures

- Given the CRM API is temporarily unavailable (e.g., returns 503 Service Unavailable)
- When the system attempts to create a ticket
- Then it must implement an exponential backoff retry strategy (up to 5 attempts over 1 hour [ASSUMED]) before alerting an administrator.

Definition of Done (DoD):

- End-to-end test from simulated call to verified CRM ticket creation succeeds.
- API payload matches the documented Swagger/OpenAPI specification.
- Retry logic is verified via mock server testing.

---

## 11. Risk Register

This section outlines the anticipated risks associated with the development, deployment, and operation of the off-hours AI Voice Agent. Risks are evaluated based on Probability (High=3, Medium=2, Low=1) and Impact (High=3, Medium=2, Low=1). The Severity Score is the product of Probability and Impact, categorized as Critical (7-9), High (4-6), Medium (2-3), or Low (1).

### 11.1 Risk Assessment Matrix

| Risk ID | Risk Description | Category | Probability | Impact | Severity Score | Mitigation Strategy | Owner |
|---|---|---|---|---|---|---|---|
| RSK-001 | Pipeline Latency Exceeds SLA: The STT -> LLM -> TTS pipeline exceeds the p99 < 800ms latency requirement, causing unnatural conversational pauses and caller abandonment. | Technical | High (3) | High (3) | 9 (Critical) | Implement streaming STT/TTS and token-streaming for the LLM. Use edge-deployed models where possible. Enforce a strict 1000ms timeout that automatically triggers a fallback to pre-recorded audio prompts. | Lead AI Engineer [ASSUMED] |
| RSK-002 | LLM Hallucination & Unauthorized Commitments: The AI agent hallucinates business policies, makes false promises, or attempts to resolve queries instead of strictly capturing data. | Business / Technical | Medium (2) | High (3) | 6 (High) | Utilize strict system prompts with few-shot examples. Implement a secondary lightweight LLM or rule-based classifier to evaluate the generated response for policy adherence before TTS synthesis. | Product Manager |
| RSK-003 | PII/PCI Data Leakage: Caller speaks sensitive personal or financial data (e.g., credit card numbers) which is transcribed and stored in plain text or sent to external LLM APIs. | Compliance / Security | Medium (2) | High (3) | 6 (High) | Integrate a real-time PII/PCI redaction service (e.g., AWS Comprehend Medical/PII [ASSUMED]) immediately post-STT. Ensure external APIs are zero-data-retention certified (e.g., OpenAI Enterprise [ASSUMED]). | Security & Compliance Officer |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| RSK-004 | Off-Hours System Downtime: The system experiences an outage during nights or weekends when IT support staffing is minimal, leading to dropped client calls. | Operational | Low (1) | High (3) | 3 (Medium) | Design a highly available, multi-region architecture. Implement an automatic, hardware-level SIP redirect to a traditional legacy voicemail system if the AI service health check fails. | DevOps Lead |
| RSK-005 | STT Failure on Edge Cases: Heavy accents, poor cellular audio quality, or loud background noise cause STT to fail, resulting in inaccurate data capture or looping "I didn't catch that" responses. | Technical | High (3) | Medium (2) | 6 (High) | Train/fine-tune STT on domain-specific and diverse accent datasets. Limit the agent to two "retry" prompts before gracefully apologizing and routing to traditional voicemail. | Data Scientist |
| RSK-006 | Prompt Injection / Malicious Callers: Callers use adversarial conversational tactics to bypass system instructions, extract system prompts, or force the agent to generate inappropriate TTS. | Security | Low (1) | Medium (2) | 2 (Medium) | Apply input sanitization and adversarial guardrails (e.g., NeMo Guardrails [ASSUMED]). Restrict the LLM's output vocabulary and enforce maximum token generation limits. | Security Engineer |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| RSK-007 | Language Detection Failure: The system fails to detect the caller's language accurately, or the caller switches languages mid-sentence, causing the TTS to respond in gibberish or the wrong language. | Technical | Medium (2) | Medium (2) | 4 (High) | Use a fast, dedicated language identification (LID) model running in parallel with STT. Default to English [ASSUMED] if confidence is < 85%. Support dynamic language switching between conversational turns, but not mid-sentence. | Lead AI Engineer |
| RSK-008 | CRM/Ticketing API Rate Limiting: High call volumes during a major off-hours event trigger CRM API rate limits, preventing the structured JSON data from being saved. | Integration | Medium (2) | High (3) | 6 (High) | Implement an asynchronous message queue (e.g., Kafka/RabbitMQ [ASSUMED]) to buffer captured call data. Add exponential backoff and retry logic for CRM API payloads. | Backend Lead |
| RSK-009 | Low User Acceptance (Immediate Hang-ups): Callers refuse to interact with an AI agent upon hearing a synthetic voice and immediately hang up, reducing the actionable data capture rate. | Business | High (3) | Medium (2) | 6 (High) | Use ultra-realistic, empathetic TTS voices (e.g., ElevenLabs [ASSUMED]). Script the opening line to immediately set expectations: "Hi, I'm the off-hours AI assistant. I can't solve your issue, but I will take your details for a priority callback tomorrow." | UX/VUI Designer |

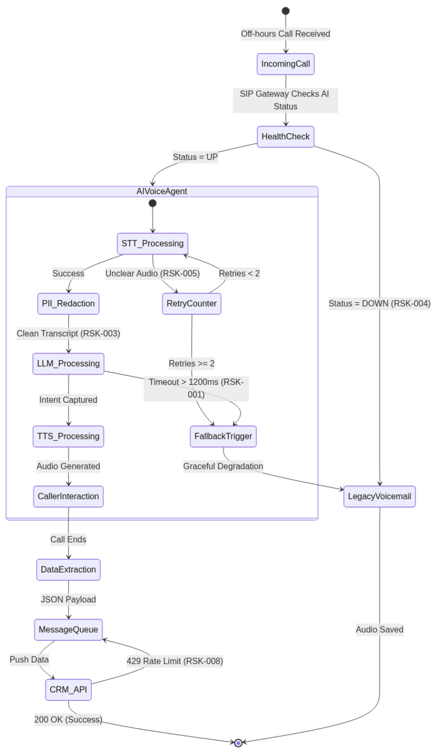
### 11.2 Risk Mitigation System Requirements

To ensure the mitigations outlined above are actionable and testable, the following system requirements are mandated.

| Req ID | Requirement Description | Priority | Acceptance Criteria |
|---|---|---|---|
| REQ-RSK-001 | Latency Fallback Mechanism: The system must monitor the STT->LLM->TTS turnaround time. If the response generation exceeds 1200ms [ASSUMED], the system must abort the AI response and play a pre-recorded audio file. | P0 - Critical | Given the LLM API is delayed,<br>When the processing time hits 1200ms,<br>Then the system plays "Please leave a message after the tone" and switches to standard voicemail mode. |
| REQ-RSK-002 | PII Redaction Pipeline: All STT transcripts must pass through a PII redaction filter before being sent to the LLM for summarization or saved to the database. | P0 - Critical | Given a caller says "My card is 4111...",<br>When the STT generates the transcript,<br>Then the transcript is mutated to "My card is [CREDIT_CARD]" before further processing. |
| REQ-RSK-003 | SIP Level Failover: The telephony gateway must ping the AI Agent health endpoint every 10 seconds [ASSUMED]. If the endpoint fails 3 consecutive pings, calls must route to legacy voicemail. | P0 - Critical | Given the AI Agent service is down,<br>When an incoming call arrives,<br>Then the SIP trunk routes the call directly to the legacy voicemail inbox without dropping the call. |
| REQ-RSK-004 | Conversational Loop Breaker: The agent must track consecutive failed intents (e.g., "I didn't understand"). After a defined threshold, it must terminate the AI interaction gracefully. | P1 - High | Given the agent fails to understand the caller twice in a row,<br>When the third failure occurs,<br>Then the agent says "I'm having trouble understanding. Please leave a voicemail," and switches to recording mode. |
| REQ-RSK-005 | Asynchronous Data Persistence: Structured call data (JSON) must be written to a resilient message queue before being pushed to the CRM. | P1 - High | Given the CRM API is returning 429 Too Many Requests,<br>When the call ends,<br>Then the JSON payload is stored in the queue and retried until successful, ensuring zero data loss. |

### 11.3 Risk Mitigation Architecture: Fallback Flow

The following Mermaid diagram illustrates the automated risk mitigation flow for technical failures (Latency, STT failure, or API downtime), ensuring the system degrades gracefully to traditional voicemail rather than dropping the client call.




## 12. Assumptions, Dependencies & Constraints

This section outlines the foundational assumptions, external dependencies, and hard constraints that dictate the boundaries of the AI-powered voice agent project. Failure to validate assumptions, secure dependencies, or adhere to constraints introduces severe risk to the project's viability, latency targets, and compliance posture.

### 12.1 Assumptions

Assumptions represent conditions we believe to be true for the purpose of planning and architecture. These must be continuously monitored and validated during the PoC and Beta phases.

| ID | Assumption Description | Priority | Validation / Acceptance Criterion |
|---|---|---|---|
| ASM-01 | Peak Concurrent Call Volume: Off-hours call volume will not exceed 500 concurrent active sessions at any given time [ASSUMED]. | High | Load testing validates system stability at 500 concurrent calls without degrading the <800ms latency target. |
| ASM-02 | Inbound Audio Quality: The telephony provider will deliver inbound SIP trunk audio using standard G.711u/a or G.722 codecs [ASSUMED] with minimal packet loss. | High | STT engine achieves a Word Error Rate (WER) of < 5% on 95% of inbound calls during the pilot phase. |
| ASM-03 | Caller AI Acceptance: Callers will engage with the AI to leave structured details rather than immediately abandoning the call upon hearing a synthetic voice. | Medium | Call abandonment rate within the first 10 seconds of AI greeting is less than 15% [ASSUMED]. |

| | | | |
|---|---|---|---|
| ASM-04 | Language Distribution: Inbound off-hours calls will be approximately 80% English, 10% Spanish, 5% French, and 5% other supported languages [ASSUMED]. | Low | Language detection logs confirm distribution; compute resources for translation are scaled accordingly without bottlenecking. |
| ASM-05 | Asynchronous CRM Processing: The downstream CRM/Ticketing system does not require synchronous confirmation back to the caller; asynchronous webhook delivery of the JSON payload is acceptable. | High | System successfully queues and delivers 100% of captured JSON payloads to the CRM within 60 seconds of call termination. |
| ASM-06 | Average Call Duration: The average interaction time to capture caller ID, intent, urgency, language, and summary will be under 120 seconds [ASSUMED]. | Medium | Telephony logs demonstrate an Average Handle Time (AHT) of "d 120 seconds across a 30-day rolling window. |

## 12.2 Dependencies

Dependencies represent external systems, third-party vendors, or cross-functional team deliverables required for the successful deployment and operation of the AI voice agent.

| ID | Dependency Description | Priority | Delivery / Acceptance Criterion |
|---|---|---|---|
| DEP-01 | Telephony / SIP Provider: Integration with Twilio/Vonage [ASSUMED] for inbound call routing, SIP trunking, and media streaming via WebSockets. | Critical | Telephony provider successfully routes inbound calls to the AI WebSocket endpoint with < 50ms network latency. |
| DEP-02 | STT & TTS Vendor SLAs: Deepgram (STT) and ElevenLabs (TTS) [ASSUMED] must provide enterprise SLAs guaranteeing high availability and ultra-low latency. | Critical | Vendor contracts guarantee 99.99% uptime and API response times of < 300ms for STT and < 250ms for TTS. |
| DEP-03 | LLM Provider SLAs: OpenAI/ Anthropic [ASSUMED] must provide provisioned throughput to ensure consistent NLU and JSON extraction without rate limiting. | Critical | LLM provider guarantees < 250ms Time to First Token (TTFT) and zero rate-limit drops during off-hours windows. |
| DEP-04 | InfoSec & Compliance Approval: The Security team must approve the PII redaction logic, data-in-transit encryption, and ephemeral storage architecture. | High | InfoSec formally signs off on the architecture review; penetration testing yields zero critical or high vulnerabilities. |
| DEP-05 | CRM API Endpoints: The internal CRM team (Salesforce/Zendesk) [ASSUMED] must provide dedicated, authenticated REST API endpoints to ingest the structured JSON payloads. | High | CRM endpoints successfully accept the defined JSON schema, returning a 201 Created status for 99.9% of valid requests. |

| | | | |
|---|---|---|---|
| DEP-06 | Voicemail Fallback Infrastructure: Existing legacy PBX/voicemail systems must remain operational to serve as the immediate failover target. | High | SIP routing rules successfully redirect calls to legacy voicemail within 2 seconds if the AI pipeline health check fails. |

### 12.3 Constraints

Constraints are hard limitations imposed on the project by business, regulatory, financial, or technical realities. The system architecture must be designed strictly within these boundaries.

| ID | Constraint Description | Priority | Compliance / Acceptance Criterion |
|---|---|---|---|
| CON-01 | Strict Latency Ceiling: The total round-trip latency (STT processing -> LLM inference -> TTS generation) must not exceed 800ms at the p99 percentile. | Critical | APM monitoring confirms p99 voice response latency remains "d 800ms over any 24-hour period. |
| CON-02 | Zero-Resolution Boundary: The AI is strictly prohibited from attempting to resolve client queries, hallucinating business policies, or making commitments. | Critical | Automated prompt testing and QA audits reveal 0% incidence of the AI offering solutions or making unauthorized commitments. |
| CON-03 | Regulatory Compliance: The system must strictly adhere to GDPR, CCPA, and SOC2 Type II [ASSUMED] requirements regarding voice biometrics and PII. | Critical | PII is successfully redacted from all stored transcripts; audio recordings are either securely encrypted or purged within 24 hours [ASSUMED]. |
| CON-04 | Operational Cost Limit: The combined API costs (Telephony + STT + LLM + TTS) must not exceed $0.15 per minute of active audio [ASSUMED]. | Medium | Monthly cloud and vendor billing reports confirm the average cost per minute of AI call handling is "d $0.15. |
| CON-05 | High Availability (Off-Hours): The system must maintain zero planned downtime during designated off-hours (Nights: 6 PM - 8 AM, Weekends: Friday 6 PM - Monday 8 AM) [ASSUMED]. | High | System achieves 99.99% uptime during defined off-hours windows, with all maintenance scheduled strictly during business hours. |
| CON-06 | Data Format Strictness: The LLM output must strictly adhere to the predefined JSON schema for CRM ingestion; malformed outputs are unacceptable. | High | 99.9% of LLM outputs pass strict JSON schema validation (Caller ID, Intent, Urgency, Language, Summary) before CRM transmission. |

## 13. Open Questions & Decision Log

This section tracks unresolved queries that carry architectural, business, or compliance implications, alongside a historical log of key architectural and product decisions made during the discovery and design phases.

### 13.1 Open Questions

The following questions must be resolved prior to the final technical design sign-off and the commencement of Sprint 1.

| ID | Open Question | Impact Assessment | Owner | Due Date | Status |
|---|---|---|---|---|---|
| OQ-001 | PII Redaction Timing: Should PII (Personally Identifiable Information) be redacted from the raw audio stream in real-time, or only from the text transcript post-call before CRM injection? | High: Real-time audio redaction adds an estimated 300-450ms latency, jeopardizing the p99 < 800ms NFR. However, storing unredacted audio poses compliance risks (GDPR/ CCPA). | Security & Compliance Lead [ASSUMED] | 2023-11-15 [ASSUMED] | In Progress |
| OQ-002 | Abusive Language Threshold: What is the exact threshold and protocol for handling abusive language? Do we issue a polite warning first, or immediately terminate the call and flag the caller ID? | Medium: Impacts brand perception and system abuse metrics. Needs alignment with existing human-agent HR policies. | Product Manager | 2023-11-12 [ASSUMED] | Open |
| OQ-003 | CRM API Rate Limits: What are the exact API rate limits for the target CRM (e.g., Salesforce/ Zendesk) during peak off-hours spikes (e.g., a major service outage at 2 AM)? | High: If the voice agent captures 500 calls in 10 minutes, synchronous CRM injection may fail. We may need an asynchronous queuing mechanism (e.g., Kafka/SQS). | Lead Architect | 2023-11-18 [ASSUMED] | Open |
| OQ-004 | Dialect & Accent Support: Will the automatic language detection default to standard locales (e.g., Standard Spanish), or must it support regional dialects (e.g., Quebecois French, Mexican Spanish) for TTS responses? | Medium: Impacts the selection of the TTS (Text-to-Speech) engine and the prompt engineering for the LLM persona. | UX/Voice Designer [ASSUMED] | 2023-11-20 [ASSUMED] | In Progress |
| OQ-005 | Silent Caller / DTMF Fallback: If a caller refuses to speak or is in a noisy environment and only uses DTMF (keypad) inputs, should the agent attempt to route them through a standard IVR tree? | Low: Edge case handling. Currently, the system assumes voice-first. DTMF fallback requires additional state machine logic. | Engineering Lead | 2023-11-14 [ASSUMED] | Open |

| | | | | | |
|---|---|---|---|---|---|
| OQ-006 | Data Retention Policy: What is the exact Time-To-Live (TTL) for the temporary S3 buckets holding the raw audio files before they are either purged or moved to cold storage (Glacier)? | High: Impacts AWS storage costs and legal discovery obligations. | Legal Counsel [ASSUMED] | 2023-11-22 [ASSUMED] | Open |

## 13.2 Decision Log

The following architectural and business decisions have been finalized. They are documented here to prevent circular discussions and to provide context for downstream engineering and QA teams.

| Decision ID | Date | Decision Made | Rationale | Alternatives Considered | Status / Approver |
|---|---|---|---|---|---|
| DEC-001 | 2023-10-15 [ASSUMED] | Strict "Capture & Route" LLM Boundary (Zero RAG) | The AI agent will not be connected to a Knowledge Base (RAG) to answer FAQs. It is strictly limited to greeting, capturing intent/ data, and closing the call. | Alternative: Allow basic FAQ resolution. Rejected due to the high risk of LLM hallucination, unauthorized commitments, and legal liability during off-hours. | Approved by VP of Product [ASSUMED] |
| DEC-002 | 2023-10-18 [ASSUMED] | Fallback to Legacy SIP Voicemail | In the event of a STT, LLM, or TTS pipeline failure (or latency exceeding 2000ms), the call will immediately bridge to the legacy SIP voicemail system. | Alternative: Play a pre-recorded error message and disconnect. Rejected because zero dropped calls is a critical business requirement. | Approved by Lead Architect [ASSUMED] |
| DEC-003 | 2023-10-22 [ASSUMED] | First 3-Second Language ID (LID) | The system will use the first 3 seconds of caller audio to detect the language, then dynamically switch the LLM prompt and TTS voice to match. | Alternative: "Press 1 for English, 2 for Spanish." Rejected as it violates the seamless, conversational UX objective. | Approved by UX Director [ASSUMED] |
| DEC-004 | 2023-10-25 [ASSUMED] | Asynchronous JSON Payload Delivery | Extracted structured data (Caller ID, Intent, Urgency, Summary) will be pushed to an SQS queue [ASSUMED] rather than synchronously written to the CRM while the caller is on the line. | Alternative: Synchronous API calls to CRM before call termination. Rejected due to the risk of CRM latency causing dead air on the voice call. | Approved by Engineering Lead [ASSUMED] |

| | | | | | |
|---|---|---|---|---|---|
| DEC-005 | 2023-10-28 [ASSUMED] | p99 Latency Target < 800ms | The entire STT -> LLM -> TTS turnaround time must not exceed 800ms at the 99th percentile. | Alternative: 1500ms target. Rejected because human conversational tolerance for pauses is ~500-800ms; anything longer causes callers to repeat themselves or hang up. | Approved by CTO [ASSUMED] |
| DEC-006 | 2023-11-02 [ASSUMED] | No Voice Biometrics for Authentication | The system will rely solely on Caller ID (ANI) and self-reported data for identification. It will not use voiceprint biometrics to authenticate callers. | Alternative: Implement voice biometrics to verify VIP clients. Rejected due to scope creep, high implementation cost, and privacy compliance complexities for Phase 1. | Approved by Security Lead [ASSUMED] |